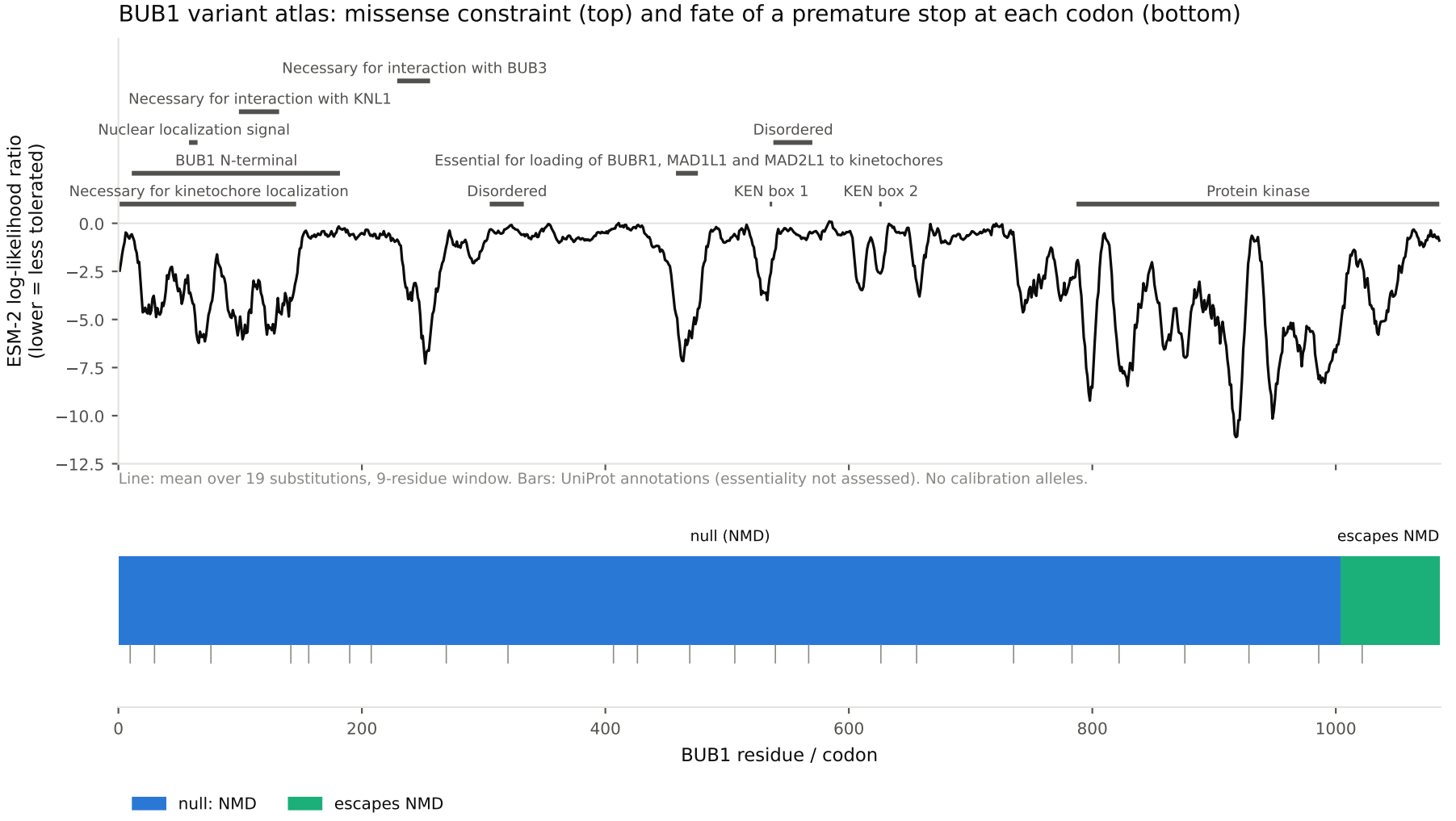
BUB1 variant atlas: missense constraint (top) and fate of a premature stop at each codon (bottom)
Necessary for interaction with BUB3
Necessary for interaction with KNL1
Nuclear localization signal
BUB1 N-terminal
Necessary for kinetochore localization
Disordered
Essential for loading of BUBR1, MAD1L1 and MAD2L1 to kinetochores
Disordered
KEN box 1
KEN box 2
Protein kinase
ESM-2 log-likelihood ratio
(lower = less tolerated)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
Line: mean over 19 substitutions, 9-residue window. Bars: UniProt annotations (essentiality not assessed). No calibration alleles.
null (NMD)
escapes NMD
0
200
400
600
800
1000
BUB1 residue / codon
null: NMD
escapes NMD